METRIC
MIL-PRF-85045/8B
W/AMENDMENT 2
13 August 2024
SUPERSEDING
MIL-PRF-85045/8B
W/AMENDMENT 1
6 August 2017

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, RADIATION HARDENED, CABLE CONFIGURATION TYPE 1 (BUFFERED FIBER), APPLICATION E (GROUND TACTICAL), CABLE CLASS MM, (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-85045.

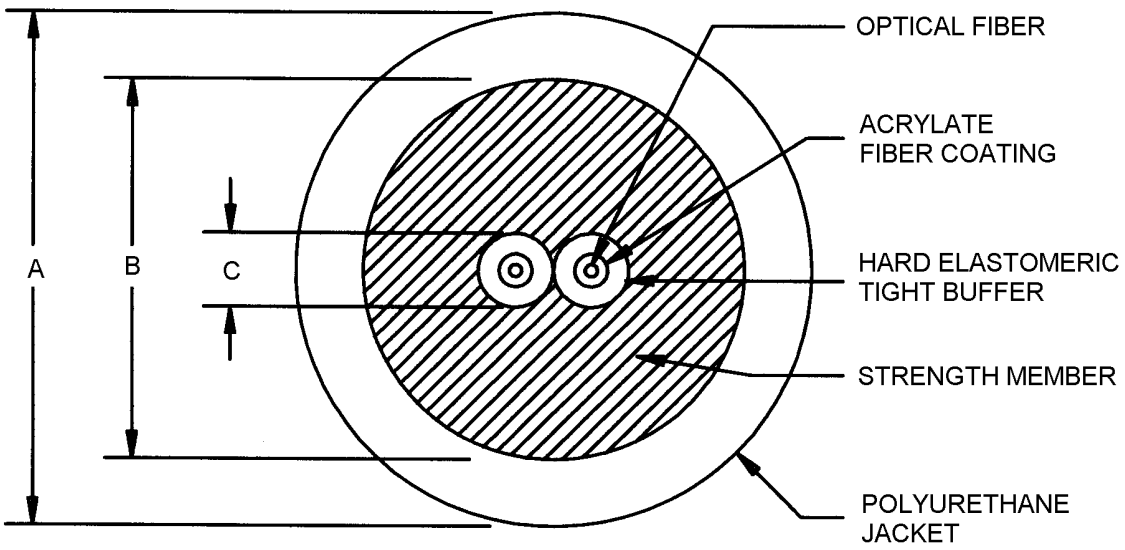


FIGURE 1. Two-fiber cable (85045/8B2X).

Dimensions		
A	B	C
5.8 ±0.15	2.7 ±0.15	0.9 ±0.05

mm	Inches
0.05	0.002
0.15	0.006
0.9	0.035
2.7	0.106
5.8	0.23

NOTE:
1. Dimensions are in millimeters.

AMSC N/A
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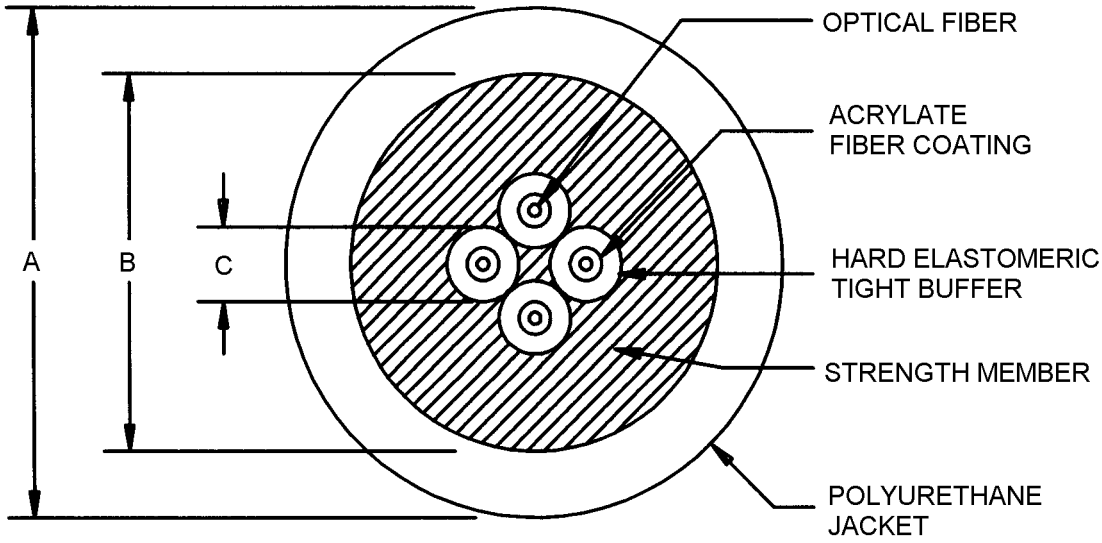


FIGURE 2. Four-fiber cable (M85045/8B4X).

Dimensions		
A	B	C
5.8 ±0.15	2.9 ±0.15	0.9 ±0.05

mm	Inches
0.05	0.002
0.15	0.006
0.9	0.035
2.9	0.114
5.8	0.23

NOTE:
1. Dimensions are in millimeters.

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CLASSIFICATION:

Fiber optic cable configuration type: 1 (buffered fiber).

Fiber cable class: MM (graded-index, glass core and glass cladding, multimode).

Part or Identifying Number (PIN): See PIN construction below. Additional manufacturer's marking is allowed.

PIN construction: M 85045 / 8 - B X X

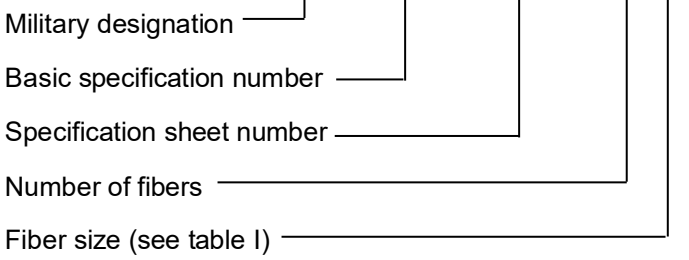


TABLE I. PIN number fiber size.

M85045/8-BXX	Fiber size (micron)
A	62.5/125
B	50/125

Example: M85045/8-B4A --- 4 fiber cable with 62.5/125 micron fiber

DESIGN AND CONSTRUCTION:

Fiber:
Class MM (50/125 μm) fibers shall be in accordance with MIL-PRF-49291/1.
Class MM (62.5/125 μm) fibers shall be in accordance with MIL-PRF-49291/6.

Buffer diameter: 0.900 ± 0.05 mm.

REQUIREMENTS:

Finished cable:

Dimensions and configuration: See figure 1 and 2.

Number of fibers: 2 or 4 (see PIN).

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Materials:

Fiber coating: Acrylate or equivalent.

Buffer: Hard elastomeric or equivalent.

Jacket material: Flame retardant polyurethane, or equivalent.

Strength members:

Polyacrylamide, type PAA (Aramid yarn), or equivalent.

Impregnated glass rods (if required).

Short term minimum bend diameter: Five times the cable outer diameter. (The short term minimum bend diameter is to be used in all environmental and mechanical tests, which specify a cable minimum bend diameter).

Long term minimum bend diameter: Ten times the cable outer diameter.

Optical fiber cable component (OFCC): Not applicable.

Cable bundle jacket: Not applicable.

Cable jacket concentricity: > 0.65.

Cable mass per unit length: 32 kg/km maximum.

PERFORMANCE REQUIREMENTS:

Optical properties:

Optical properties inspections: The optical requirements specified herein shall be met at the center wavelength of 1300 nm with a plus or minus 20 nm tolerance.

Attenuation rate: The multimode fiber attenuation rate shall be measured in accordance with TIA-455-78. For multimode fibers, the source shall be noncoherent. Light launch conditions used during the attenuation rate measurements shall be made in accordance with TIA-455-78 and Table II. The attenuation rate of each individual single mode fiber shall be measured in accordance with TIA-455-78.

TABLE II. Light Launch conditions for attenuation rate tests.

Fiber Type	Launch conditions
MM	70/70 restricted or equivalent

Change in optical transmittance: This test shall evaluate the change in optical transmittance of the fibers due to exposure of the cable to one or more inspection (environment and physical) tests.

Method: The change in optical transmittance of each fiber shall be measured in accordance with TIA-455-20, utilizing a monitor fiber to evaluate the change in transmittance due to exposure of the cable to environmental and physical tests. Any optical power detection method may be utilized if the method has the sensitivity to measure the differential optical power levels as specified in the individual mechanical test requirements of section 3, and if the method provides repeatable readings (less than 3 percent variation). A pretest optical power measurement shall be made and the specimen shall then undergo inspection testing. All optical power measurements, after the pretest measurement, shall be referenced to the pretest value and the change in dB calculated.

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Guidelines: These types of measurements require highly stable optical devices (source and detector) and repeatability of loss at the device-to-fiber interface. Use of the same reference fiber for calibrating the light source power output just prior to making all the measurements on the cable specimen, will enhance the measurement accuracy.

Crosstalk (see 3.5.3): Fully assembled cable specimens shall be tested for crosstalk in accordance with TIA-455-42. When applicable, the crosstalk shall be determined between a minimum of three randomly selected fibers in a multi-fiber cable, using each selected fiber sequentially as the actively transmitting element, and measuring the far-end power output of all other fibers in the cable specimen. The center wavelength shall be 1300 nm ±20 nm, and the light launch conditions shall be as specified (see Table II).

Maximum attenuation rate:

50/125 micron fiber: 3.50 dB/km at 850 nm ± 20 nm.
1.0 dB/km at 1,300 nm ± 20 nm.

62.5/125 micron fiber: 3.75 dB/km at 850 nm ± 20 nm.
1.50 dB/km at 1,300 nm ± 20 nm.

End resultant attenuation due to cumulative environmental and mechanical testing: Not applicable.

Mechanical properties:

Tensile loading and elongation: Applicable, except the induced attenuation shall be measured only after releasing the load. The complete post-test visual and mechanical testing is not required. A post-test visual jacket examination shall be made under 10X magnification.

Operating tensile load: Applicable, except the complete post-test visual and mechanical testing is not required. A post-test visual jacket examination shall be made under 10X magnification.

Dynamic bend: Not applicable.

Corner bend: Applicable, except the test force shall be 500 N.

Cable jacket material tensile strength and elongation: Applicable.

Durability of identification marking: Applicable, except the applied mass of 100 grams shall be applied.

Environmental:

Temperature range:
Operating temperature: -46°C, +71°C.
Storage temperature: -57°C, +85°C.

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Thermal shock: Use test schedule A.

Fluid Immersion: Use JP-8 conforming to MIL-DTL-83133 may be used in lieu of AMS2629.

Barometric pressure (reduced):
Operating: 3,000 m. (Test condition letter B).
Nonoperating: 12,200 m.

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Life aging: Test exposure time shall be 240 hours at 110°C.

Freezing water immersion: Applicable. Use method A, procedure I.

Flame extinguishing and smoke generation: Not applicable.

Smoke generation and flame propagation: Not applicable.

Acid gas generation: Not applicable.

Shock: Not applicable.

Water absorption: Applicable except the sample size shall be in accordance with ASTM D470 and test shall be performed in accordance with the gravimetric method of ASTM D470.

Paint susceptibility: Not applicable.

Halogen content: Not applicable.

Toxicity: Not applicable.

QUALIFICATION BY SIMILARITY:

Qualified 50/125 2 fiber cable and candidate 62.5/125 2 fiber cable:

Manufacturers who are qualified under this specification sheet for 50/125 micron 2 fiber cable and whose 62.5/125 micron 2 fiber cable passes the following tests and inspections specified in Table III are qualified by similarity under this specification. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber had been changed from 50/125 to 62.5/125 optical fiber. Testing and inspection may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE III Qualification by similarity (fiber size).

Group I
visual and mechanical
attenuation rate
Group III
temperature cycling
temperature/humidity cycling
storage temperature
cyclic flexing
crush
cable twist-bend
impact, (low temperature only)
tensile loading and elongation
operating tensile loading
Group IV
thermal shock

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Qualified 50/125 4 fiber cable and produce 50/125 2 fiber cable:

Manufacturers who are qualified under this specification sheet for 50/125 micron 4 fiber cable and whose 50/125 micron 2 fiber cable passes the following tests and inspections specified in Table IV are qualified by similarity under this specification. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the number of fibers in the cable had been changed from 4 fibers to 2 fibers. Testing and inspection may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE IV Qualification by similarity (fiber number).

Group I
visual and mechanical
Group III
cyclic flexing
crush
cable twist-bend
impact, (low temperature only)
tensile loading and elongation
operating tensile loading

Qualified 50/125 2 and 4 fiber cables and candidate 62.5 2 and 4 fiber cables:

Manufacturers who are qualified under this specification sheet for 50/125 micron 2 and 4 fiber cable and whose 62.5/125 micron fiber cable passes the tests and inspections specified in Table V are qualified by similarity under this specification. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber had been changed from 50/125 to 62.5/125 optical fiber. Testing and inspection may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

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TABLE V Qualification by similarity (fiber size and number).

Group I
visual and mechanical
attenuation rate
Group III
temperature cycling
temperature/humidity cycling
storage temperature
cyclic flexing
crush
cable twist-bend
impact, (low temperature only)
tensile loading and elongation
operating tensile loading
Group IV
thermal shock

Qualified 62.5/125 2 fiber cable and candidate 50/125 2 fiber cable:

Manufacturers who are qualified under this specification sheet for 62.5/125 micron 2 fiber cable and whose 50/125 micron 2 fiber cable passes the tests and inspections specified in Table VI are qualified by similarity under this specification. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber had been changed from 62.5/125 to 50/125 optical fiber. Testing and inspection may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

TABLE VI Qualification by similarity (fiber size).

Group I
visual and mechanical
attenuation rate
Group III
temperature cycling
temperature/humidity cycling
storage temperature
cyclic flexing
crush
cable twist-bend
impact, (low temperature only)
tensile loading and elongation
operating tensile loading
Group IV
thermal shock

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-85045, this document references the following:

MIL-PRF-49291/1 ASTM D470 TIA-455-20 TIA-455-78

MIL-PRF-49291/6 TIA-455-42 MIL-DTL-83133 AMS2629

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project number: 6015-2024-004)

Review activities:
Navy - AS
Air Force – 13, 19, 93
DIA – DI

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